

INTERVAL ESTIMATION

Confidence Interval

A Confidence Interval (CI) is a range of values that's likely to include a population value with a certain degree of confidence. It is often expressed as a percentage whereby a population mean, or proportion, lies between an upper and lower interval.

Critical value for confidence interval

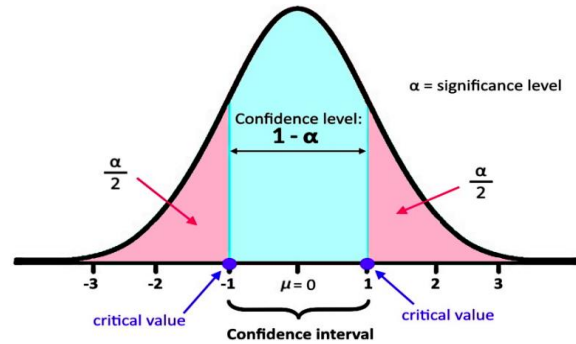
Consider α is the level of significance, and hence $1 - \alpha$ is the level of confidence, to find the critical value for the confidence interval $\pm z_{\alpha/2}$. Then, we need to assume

$$P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha$$

$$\text{or, } 2\phi(z_{\alpha/2}) - 1 = 1 - \alpha$$

$$\text{or, } 2\phi(z_{\alpha/2}) = 2 - \alpha$$

$$\text{or, } \phi(z_{\alpha/2}) = \frac{2 - \alpha}{2}$$



Confidence intervals for the mean (μ)

Assume n is the size of the sample, $\bar{X}$ is the average of the sample data, σ is the standard deviation of the target data, and μ is the required mean of the population. To find the required confidence interval, we need to assume

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Then the confidence interval for the mean is

$$-z_{\alpha/2} \leq Z \leq z_{\alpha/2}$$

$$\text{or, } -z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \leq z_{\alpha/2}$$

$$\text{or, } -z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) \leq \bar{X} - \mu \leq z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

$$\text{or, } -\bar{X} - z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) \leq -\mu \leq -\bar{X} + z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

$$\text{or, } \bar{X} + z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) \geq \mu \geq \bar{X} - z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

$$\text{or, } \bar{X} - z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) \leq \mu \leq \bar{X} + z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

INTERVAL ESTIMATION

Confidence intervals for the proportion (p)

Consider, n is the size of the target samples, Y is the number of sample elements in favor of an event, and p is the required proportion of the population. To find the required confidence interval, we need to assume

$$Z = \frac{Y - \mu}{\sigma} = \frac{Y - np}{\sqrt{np(1-p)}} = \frac{\frac{Y}{n} - p}{\sqrt{\frac{Y}{n}\left(1 - \frac{Y}{n}\right)}} \cdot \sqrt{\frac{n}{n}}$$

Then the confidence interval for the proportion is

$$-z_{\alpha/2} \leq Z \leq z_{\alpha/2}$$

$$\text{or, } -z_{\alpha/2} \leq \frac{\frac{Y}{n} - p}{\sqrt{\frac{Y}{n}\left(1 - \frac{Y}{n}\right)}} \leq z_{\alpha/2}$$

$$\text{or, } -z_{\alpha/2} \sqrt{\frac{Y}{n}\left(1 - \frac{Y}{n}\right)} \leq \frac{Y}{n} - p \leq z_{\alpha/2} \sqrt{\frac{Y}{n}\left(1 - \frac{Y}{n}\right)}$$

$$\text{or, } -\frac{Y}{n} - z_{\alpha/2} \sqrt{\frac{Y}{n}\left(1 - \frac{Y}{n}\right)} \leq -p \leq -\frac{Y}{n} + z_{\alpha/2} \sqrt{\frac{Y}{n}\left(1 - \frac{Y}{n}\right)}$$

$$\text{or, } \frac{Y}{n} + z_{\alpha/2} \sqrt{\frac{Y}{n}\left(1 - \frac{Y}{n}\right)} \geq p \geq \frac{Y}{n} - z_{\alpha/2} \sqrt{\frac{Y}{n}\left(1 - \frac{Y}{n}\right)}$$

$$\text{or, } \frac{Y}{n} - z_{\alpha/2} \sqrt{\frac{Y}{n}\left(1 - \frac{Y}{n}\right)} \leq p \leq \frac{Y}{n} + z_{\alpha/2} \sqrt{\frac{Y}{n}\left(1 - \frac{Y}{n}\right)}$$

1. Let X equal the amount of food (in pounds per day) consumed by a worker having the standard deviation $\sigma = 1.2$. To estimate the mean μ of X , a random sample of 50 workers has been taken, and found that they consumed a total of 230 lb of food per day. Find the confidence interval for μ with a 5% significance level.

Solution: Given by

Sample size $n = 50$

Mean consumption $\bar{X} = \frac{230}{50} = 4.6$ pounds

Standard deviation $\sigma = 1.2$ pounds

Significance $\alpha = 0.05$

Critical value: $\phi(z_{\alpha/2}) = \frac{2-\alpha}{2} = \frac{2-0.05}{2} = 0.975 \rightarrow z_{\alpha/2} = 1.96$

INTERVAL ESTIMATION

Now, the required confidence interval is

$$4.6 - 1.96 \left(\frac{1.2}{\sqrt{50}} \right) \leq \mu \leq 4.6 + 1.96 \left(\frac{1.2}{\sqrt{50}} \right)$$

$$\text{or, } 4.6 - 0.33 \leq \mu \leq 4.6 + 0.33$$

$$\text{or, } 4.27 \leq \mu \leq 4.93$$

2. Let X equal the amount of food (in pounds per day) consumed by a student. Suppose the variance σ^2 of X is 1.21. To estimate the mean μ of X , an agency took a random sample of 1000 people and found that they consumed in total 4200 lb of food per day. Find an approximate 95% confidence interval for μ .

Solution: Given by

$$\text{Sample size } n = 1000$$

$$\text{Mean consumption } \bar{X} = \frac{4200}{1000} = 4.2 \text{ lb}$$

$$\text{Variance } \sigma^2 = 1.21 \rightarrow \text{Standard deviation } \sigma = 1.1 \text{ lb}$$

$$\text{Confidence } 1 - \alpha = 0.95 \rightarrow \text{Significance } \alpha = 0.05$$

$$\text{Critical value: } \phi(z_{\alpha/2}) = \frac{2-\alpha}{2} = \frac{2-0.05}{2} = 0.975 \rightarrow z_{\alpha/2} = 1.96$$

Now, the required confidence interval is

$$4.2 - 1.96 \left(\frac{1.1}{\sqrt{1000}} \right) \leq \mu \leq 4.2 + 1.96 \left(\frac{1.1}{\sqrt{1000}} \right)$$

$$\text{or, } 4.2 - 0.07 \leq \mu \leq 4.2 + 0.07$$

$$\text{or, } 4.13 \leq \mu \leq 4.27$$

3. Let X equal the weight of fruits in kg per day consumed by a student. Suppose the standard deviation of X is 0.1 kg. To estimate the mean μ of X , an agency took a random sample of 20 students and found they consumed 10 kg of fruits per day. Find an approximate 90% confidence interval for μ .

Solution: Given by

$$\text{Sample size } n = 20$$

$$\text{Mean consumption } \bar{X} = \frac{10}{20} = 0.5 \text{ kg}$$

$$\text{Standard deviation } \sigma = 0.1 \text{ kg}$$

$$\text{Confidence } 1 - \alpha = 0.90 \rightarrow \text{Significance } \alpha = 0.10$$

INTERVAL ESTIMATION

$$\text{Critical value: } \phi(z_{\alpha/2}) = \frac{2-\alpha}{2} = \frac{2-0.10}{2} = 0.95 \rightarrow z_{\alpha/2} = 1.64$$

Now, the required confidence interval is

$$\begin{aligned} 0.5 - 1.64 \left(\frac{0.1}{\sqrt{20}} \right) &\leq \mu \leq 0.5 + 1.64 \left(\frac{0.1}{\sqrt{20}} \right) \\ \text{or, } 0.5 - 0.04 &\leq \mu \leq 0.5 + 0.04 \\ \text{or, } 0.46 &\leq \mu \leq 0.54 \end{aligned}$$

4. Let X equal the amount of juice in milliliters per day consumed by a tourist. Suppose the variance of X is 36. To estimate the mean μ of X , a survey team took a random sample of 50 tourists and found they consumed on average 0.5 liter juice per day. Find an approximate 90% confidence interval for μ .

Solution: Given by

$$\text{Sample size } n = 50$$

$$\text{Mean consumption } \bar{X} = 0.5 \text{ liter or } \bar{X} = 500 \text{ milliliter}$$

$$\text{Variance } \sigma^2 = 36 \rightarrow \text{Standard deviation } \sigma = 6 \text{ milliliter}$$

$$\text{Confidence } 1 - \alpha = 0.90 \rightarrow \text{Significance } \alpha = 0.10$$

$$\text{Critical value: } \phi(z_{\alpha/2}) = \frac{2-\alpha}{2} = \frac{2-0.10}{2} = 0.95 \rightarrow z_{\alpha/2} = 1.64$$

Now, the required confidence interval is

$$\begin{aligned} 500 - 1.64 \left(\frac{6}{\sqrt{50}} \right) &\leq \mu \leq 500 + 1.64 \left(\frac{6}{\sqrt{50}} \right) \\ \text{or, } 500 - 1.39 &\leq \mu \leq 500 + 1.39 \\ \text{or, } 498.61 &\leq \mu \leq 501.39 \end{aligned}$$

5. Let X equal the amount of coffee (in milliliters per day) consumed by a man. Suppose the variance σ^2 of X is 32. To estimate the mean μ of X , an agency took a random sample of 10 men and found they consumed on average 150 milliliters of coffee per day. Find an approximate 80% confidence interval for μ .

Solution: Given by

$$\text{Sample size } n = 10$$

$$\text{Mean consumption } \bar{X} = 150 \text{ milliliter}$$

INTERVAL ESTIMATION

Variance $\sigma^2 = 32 \rightarrow$ Standard deviation $\sigma = \sqrt{32}$ milliliter

Confidence $1 - \alpha = 0.80 \rightarrow$ Significance $\alpha = 0.20$

Critical value: $\phi(z_{\alpha/2}) = \frac{2-\alpha}{2} = \frac{2-0.20}{2} = 0.90 \rightarrow z_{\alpha/2} = 1.28$

Now, the required confidence interval is

$$150 - 1.28 \left(\frac{\sqrt{32}}{\sqrt{10}} \right) \leq \mu \leq 150 + 1.28 \left(\frac{\sqrt{32}}{\sqrt{10}} \right)$$

$$\text{or, } 150 - 2.29 \leq \mu \leq 150 + 2.29$$

$$\text{or, } 147.71 \leq \mu \leq 152.29$$

6. A builder's merchant sells stones of different sizes. The masses of M-sized stones have a standard deviation of 6 grams. The mean mass of a random sample of 200 M-sized stones is 45 grams. Find a 95% confidence interval for the mean mass of M-sized stones.

Solution: Given by

Sample size $n = 200$

Mean mass $\bar{X} = 45$ gram

Standard deviation $\sigma = 6$ gram

Confidence $1 - \alpha = 0.95 \rightarrow$ Significance $\alpha = 0.05$

Critical value: $\phi(z_{\alpha/2}) = \frac{2-\alpha}{2} = \frac{2-0.05}{2} = 0.975 \rightarrow z_{\alpha/2} = 1.96$

Now, the required confidence interval is

$$45 - 1.96 \left(\frac{6}{\sqrt{200}} \right) \leq \mu \leq 45 + 1.96 \left(\frac{6}{\sqrt{200}} \right)$$

$$\text{or, } 45 - 0.83 \leq \mu \leq 45 + 0.83$$

$$\text{or, } 44.17 \leq \mu \leq 45.83$$

7. In the forest, there were 800 plants under extinction, 70% of the plants were transferred for saving from extinction. If 75% of the transferred plants exist after the attempt, find the confidence interval of the proportion with a 10% significance level.

Solution: Given by

Target sample size $n = 70\% \text{ of } 800 = 560$

INTERVAL ESTIMATION

Number of favorable samples $Y = 75\% \text{ of } 560 = 420$

Probability of success in the samples $\frac{Y}{n} = 0.75$

Significance $\alpha = 0.10$

Critical value: $\phi(z_{\alpha/2}) = \frac{2-\alpha}{2} = \frac{2-0.10}{2} = 0.95 \rightarrow z_{\alpha/2} = 1.64$

Now, the required confidence interval is

$$0.75 - 1.64 \left(\sqrt{\frac{0.75(1-0.75)}{560}} \right) \leq p \leq 0.75 + 1.64 \left(\sqrt{\frac{0.75(1-0.75)}{560}} \right)$$

$$\text{or, } 0.75 - 0.03 \leq p \leq 0.75 + 0.03$$

$$\text{or, } 0.72 \leq p \leq 0.78$$

8. In a certain pollution awareness campaign, one advisor conducted a survey taken at random among 300 people, with 65% of them providing their opinion about the awareness. If the survey secured 170 positive opinions. Find an approximate 95% confidence interval for the fraction p of the people who support the advisor.

Solution: Given by

Target sample size $n = 65\% \text{ of } 300 = 195$

Number of favorable samples $Y = 170$

Probability of success in the samples $\frac{Y}{n} = 0.872$

Confidence $1 - \alpha = 0.95 \rightarrow$ Significance $\alpha = 0.05$

Critical value: $\phi(z_{\alpha/2}) = \frac{2-\alpha}{2} = \frac{2-0.05}{2} = 0.975 \rightarrow z_{\alpha/2} = 1.96$

Now, the required confidence interval is

$$0.872 - 1.96 \left(\sqrt{\frac{0.872(1-0.872)}{195}} \right) \leq p \leq 0.872 + 1.96 \left(\sqrt{\frac{0.872(1-0.872)}{195}} \right)$$

$$\text{or, } 0.872 - 0.047 \leq p \leq 0.872 + 0.047$$

$$\text{or, } 0.825 \leq p \leq 0.919$$

INTERVAL ESTIMATION

9. In a certain zoo, there were 1200 animals under severe viral infection, 85% of the animals were transferred to another zoo. If half of the total animals survived after the attempt, find the confidence interval of the proportion with a 1% significance level.

Solution: Given by

$$\text{Target sample size } n = 85\% \text{ of } 1200 = 1020$$

$$\text{Number of favorable samples } Y = 50\% \text{ of } 1200 = 600$$

$$\text{Probability of success in the samples } \frac{Y}{n} = 0.588$$

$$\text{Significance } \alpha = 0.01$$

$$\text{Critical value: } \phi(z_{\alpha/2}) = \frac{2-\alpha}{2} = \frac{2-0.01}{2} = 0.995 \rightarrow z_{\alpha/2} = 2.58$$

Now, the required confidence interval is

$$0.588 - 2.58 \left(\sqrt{\frac{0.588(1-0.588)}{1020}} \right) \leq p \leq 0.588 + 2.58 \left(\sqrt{\frac{0.588(1-0.588)}{1020}} \right)$$

$$\text{or, } 0.588 - 0.040 \leq p \leq 0.588 + 0.040$$

$$\text{or, } 0.548 \leq p \leq 0.628$$

10. In a forest, there were 200 birds in severe habitat trouble, 75% of the birds were rescued from the forest. If 80% of the rescued birds survived after the attempt, find the confidence interval of the proportion with an 85% confidence level.

Solution: Given by

$$\text{Target sample size } n = 75\% \text{ of } 200 = 150$$

$$\text{Number of favorable samples } Y = 80\% \text{ of } 150 = 120$$

$$\text{Probability of success in the samples } \frac{Y}{n} = 0.8$$

$$\text{Confidence } 1 - \alpha = 0.85 \rightarrow \text{Significance } \alpha = 0.15$$

$$\text{Critical value: } \phi(z_{\alpha/2}) = \frac{2-\alpha}{2} = \frac{2-0.15}{2} = 0.925 \rightarrow z_{\alpha/2} = 1.44$$

Now, the required confidence interval is

$$0.8 - 1.44 \left(\sqrt{\frac{0.8(1-0.8)}{150}} \right) \leq p \leq 0.8 + 1.44 \left(\sqrt{\frac{0.8(1-0.8)}{150}} \right)$$

INTERVAL ESTIMATION

$$\text{or, } 0.8 - 0.047 \leq p \leq 0.8 + 0.047$$

$$\text{or, } 0.753 \leq p \leq 0.847$$

11. In a certain political campaign, one candidate has a poll taken at random among 2500 people, with 60% of them being voters. If the candidate secured 55% of casted votes. Find an approximate 95% confidence interval for the fraction p of the voting population that favors the candidate.

Solution: Given by

$$\text{Target sample size } n = 60\% \text{ of } 2500 = 1500$$

$$\text{Number of favorable samples } Y = 55\% \text{ of } 1500 = 825$$

$$\text{Probability of success in the samples } \frac{Y}{n} = 0.55$$

$$\text{Confidence } 1 - \alpha = 0.95 \rightarrow \text{Significance } \alpha = 0.05$$

$$\text{Critical value: } \phi(z_{\alpha/2}) = \frac{2-\alpha}{2} = \frac{2-0.05}{2} = 0.975 \rightarrow z_{\alpha/2} = 1.96$$

Now, the required confidence interval is

$$0.55 - 1.96 \left(\sqrt{\frac{0.55(1-0.55)}{1500}} \right) \leq p \leq 0.55 + 1.96 \left(\sqrt{\frac{0.55(1-0.55)}{1500}} \right)$$

$$\text{or, } 0.55 - 0.025 \leq p \leq 0.55 + 0.025$$

$$\text{or, } 0.525 \leq p \leq 0.575$$

12. A random sample of 200 students is selected, and 126 of them report that they prefer online classes. Construct a 95% confidence interval for the actual population proportion.

Solution: Given by

$$\text{Target sample size } n = 200$$

$$\text{Number of favorable samples } Y = 126$$

$$\text{Probability of success in the samples } \frac{Y}{n} = 0.63$$

$$\text{Confidence } 1 - \alpha = 0.95 \rightarrow \text{Significance } \alpha = 0.05$$

$$\text{Critical value: } \phi(z_{\alpha/2}) = \frac{2-\alpha}{2} = \frac{2-0.05}{2} = 0.975 \rightarrow z_{\alpha/2} = 1.96$$

Now, the required confidence interval is

INTERVAL ESTIMATION

$$0.63 - 1.96 \left(\sqrt{\frac{0.63(1-0.63)}{1020}} \right) \leq p \leq 0.63 + 1.96 \left(\sqrt{\frac{0.63(1-0.63)}{1020}} \right)$$

$$\text{or, } 0.63 - 0.07 \leq p \leq 0.63 + 0.07$$

$$\text{or, } 0.56 \leq p \leq 0.70$$